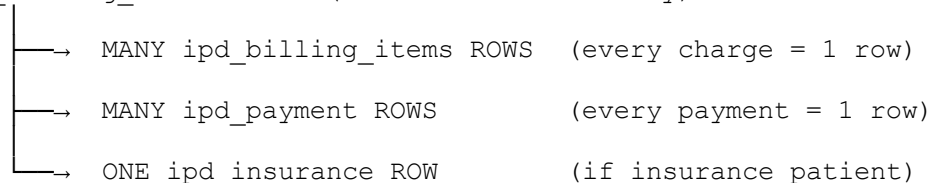


```
# IPD Billing Module – Final Plan (Option A)
### Row-per-Charge Architecture | Master-Detail Design
---
```

1. THE BIG PICTURE

```
...
```

ONE PATIENT → ONE ipd_billing_master ROW (the financial summary)



```
...
```

Golden Rules:

- ☒ 1 master per admission – enforced in PHP logic
- ☒ Every charge = 1 row in items – never combined
- ☒ Every payment = 1 row in payments – never overwritten
- ☒ All totals calculated server-side – frontend only displays
- ☒ Nothing is ever deleted – only cancelled (status flag)

```
---
```

2. DATABASE – WHAT CHANGES

> No new tables needed. Only ALTER TABLE on 4 existing tables.

```
---
```

2.1 `ipd\_billing\_master` – ADD 7 Columns

Current missing columns (need to add):

```
```sql
```

```
ALTER TABLE ipd_billing_master
 ADD COLUMN doctor_charges DECIMAL(10,2) DEFAULT 0.00 AFTER room_charges,
 ADD COLUMN lab_charges DECIMAL(10,2) DEFAULT 0.00 AFTER doctor_charges,
 ADD COLUMN radiology_charges DECIMAL(10,2) DEFAULT 0.00 AFTER lab_charges,
 ADD COLUMN pharmacy_charges DECIMAL(10,2) DEFAULT 0.00 AFTER radiology_charges,
 ADD COLUMN ot_charges DECIMAL(10,2) DEFAULT 0.00 AFTER pharmacy_charges,
 ADD COLUMN procedure_charges DECIMAL(10,2) DEFAULT 0.00 AFTER ot_charges,
 ADD COLUMN updated_by VARCHAR(50) NULL AFTER created_by;
```

```
```
```

What each column stores after adding:

| Column | What It Stores | How It Gets Populated |
|------------------|--|--------------------------|
| `bill_id` | Unique bill ID (e.g. BILL-20250726-0001) | Auto-generated on create |
| `admission_id` | FK to ipd_admissions | Set on create |
| `patient_id` | FK to patient | Set on create |
| `doctor_id` | Admitting doctor | Set on create |
| `admission_date` | Date of admission | Set on create |
| `discharge_date` | Date of discharge | Updated on discharge |
| `total_days` | Days stayed | Recalculated |
| `room_charges` | SUM of ROOM_RENT items | `recalculateMaster()` |

```

`doctor_charges` *(NEW)*	SUM of DOCTOR_VISIT items	`recalculateMaster()`
`lab_charges` *(NEW)*	SUM of LAB items	`recalculateMaster()`
`radiology_charges` *(NEW)*	SUM of RADIOLOGY items	`recalculateMaster()`
`pharmacy_charges` *(NEW)*	SUM of PHARMACY items	`recalculateMaster()`
`ot_charges` *(NEW)*	SUM of OT items	`recalculateMaster()`
`procedure_charges` *(NEW)*	SUM of PROCEDURE items	`recalculateMaster()`
`consumable_charges`	SUM of CONSUMABLE items	`recalculateMaster()`
`other_charges`	SUM of OTHER items	`recalculateMaster()`
`subtotal`	SUM of all 9 category charges	`recalculateMaster()`
`discount_amount`	Fixed discount in ₹	Manually set
`discount_percentage`	Discount %	Manually set
`grand_total`	subtotal - discount_amount	`recalculateMaster()`
`bill_type`	SELF / INSURANCE / CORPORATE	Set on create or update
`insurance_company_id`	Insurance company ref	Set for insurance patients
`policy_number`	Policy number	Set for insurance patients
`approval_number`	Insurance approval number	Set when approved
`insurance_approved_amount`	Amount approved by insurer	From ipd_insurance
`insurance_received_amount`	Amount actually received	SUM from ipd_payment
(is_insurance=1)		
`patient_payable`	grand_total - insurance_approved_amount	
`recalculateMaster()`		
`advance_amount`	SUM of ADVANCE payments	`recalculateMaster()`
`amount_paid`	SUM of all patient payments	`recalculateMaster()`
`balance_due`	grand_total - amount_paid - insurance_received	
`recalculateMaster()`		
`payment_mode`	Last payment mode	Updated on each payment
`payment_status`	Pending / Partial / Paid / Cancelled	`recalculateMaster()`
`billing_status`	OPEN / UNDER_TREATMENT / DISCHARGE_PENDING / FINALIZED /	
CANCELLED	Manually changed	
`referral_type`	Internal / External	Set on create
`referred_by`	Referring person name	Set on create
`sponsor`	Sponsor name	Set on create
`notes`	General notes	Set anytime
`created_by`	User who created	Set on create
`updated_by` *(NEW)*	User who last updated	Updated on each change
`created_at`	Creation timestamp	Auto
`updated_at`	Last update timestamp	Auto

```

2.2 `ipd\_billing\_items` - ADD 6 Columns

```

```sql
ALTER TABLE ipd_billing_items
 ADD COLUMN quantity DECIMAL(8,2) DEFAULT 1.00 AFTER description,
 ADD COLUMN unit_price DECIMAL(10,2) DEFAULT 0.00 AFTER quantity,
 ADD COLUMN discount_amt DECIMAL(10,2) DEFAULT 0.00 AFTER unit_price,
 ADD COLUMN bed_rent DECIMAL(10,2) DEFAULT 0.00 AFTER discount_amt,
 ADD COLUMN nursing_charge DECIMAL(10,2) DEFAULT 0.00 AFTER bed_rent,
 ADD COLUMN duty_dr_charge DECIMAL(10,2) DEFAULT 0.00 AFTER nursing_charge;
```

```

What each row looks like for each charge type:

...

| | |
|--------------------------------------|-----------|
| ROOM RENT ROW (auto-generated daily) | |
| charge_type | ROOM_RENT |

| | |
|-------------|--|
| charge_date | 2025-07-20 |
| description | Room Rent - General Ward - Bed B-05 - 20-Jul-2025 |
| quantity | 1 |
| unit_price | 2000 (total_bed_amount from hospital_beds) |
| bed_rent | 1500 (amount_per_day from hospital_beds) |
| nursing_ch | 300 (nursig_charge from hospital_beds - typo column) |
| duty_dr_ch | 200 (doctor_charge from hospital_beds) |
| discount_a | 0 |
| total_amou | 2000 |
| items_json | {"ward":"General Ward","bed":"B-05","room":"101"} |
| status | COMPLETED |

| | |
|-------------------------------------|--|
| DOCTOR VISIT ROW (manually entered) | |
| charge_type | DOCTOR_VISIT |
| charge_date | 2025-07-22 |
| description | Cardiologist Visit - Dr. Sharma |
| quantity | 1 |
| unit_price | 1500 |
| bed_rent | 0 (not applicable) |
| nursing_ch | 0 (not applicable) |
| duty_dr_ch | 0 (not applicable) |
| discount_a | 0 |
| total_amou | 1500 |
| items_json | {"doctor_id":"D005","specialization":"Cardiology"} |
| status | COMPLETED |

| | |
|---|---|
| LAB ROW (manually entered - multiple tests = multiple rows) | |
| charge_type | LAB |
| charge_date | 2025-07-21 |
| description | CBC - Complete Blood Count |
| quantity | 1 |
| unit_price | 500 |
| total_amou | 500 |
| items_json | {"test_code":"CBC","sample":"Blood","lab_ref":"L001"} |

| | |
|---|--------------------------------------|
| PHARMACY ROW (each medicine = 1 row OR group for same date) | |
| charge_type | PHARMACY |
| charge_date | 2025-07-23 |
| description | Paracetamol 500mg |
| quantity | 10 (tablets) |
| unit_price | 5 (per tablet) |
| total_amou | 50 |
| items_json | {"batch":"B2025","expiry":"2026-01"} |

...

2.3 `ipd\_payment` - ADD 1 Column

```

```sql
ALTER TABLE ipd_payment
 ADD COLUMN is_insurance TINYINT(1) DEFAULT 0
 COMMENT '0=patient payment, 1=insurance payout'
 AFTER reference_no;
```

```

****What each payment row looks like:****

...

Patient Payment (Advance):

| | |
|--------------|---------------------|
| bill_id | BILL-20250726-0001 |
| payment_type | ADVANCE |
| payment_mode | CASH |
| amount | 5000.00 |
| is_insurance | 0 ← patient payment |
| reference_no | NULL |

Patient Payment (Partial via UPI):

| | |
|--------------|--------------------|
| bill_id | BILL-20250726-0001 |
| payment_type | PARTIAL |
| payment_mode | UPI |
| amount | 3000.00 |
| is_insurance | 0 |
| reference_no | UPI-TXN-123456 |

Insurance Payout (received from insurer):

| | |
|--------------|-----------------------|
| bill_id | BILL-20250726-0001 |
| payment_type | FINAL |
| payment_mode | INSURANCE |
| amount | 60000.00 |
| is_insurance | 1 ← insurance payment |
| reference_no | INS-CLAIM-78901 |

Refund:

| | |
|--------------|----------------------------------|
| payment_type | REFUND |
| amount | 500.00 ← positive (not negative) |
| is_insurance | 0 |

...

2.4 `ipd\_insurance` – ADD 5 Columns

```

```sql
ALTER TABLE ipd_insurance
 ADD COLUMN insurance_type ENUM('INSURANCE','CORPORATE') DEFAULT 'INSURANCE'
 AFTER patient_id,
 ADD COLUMN company_name VARCHAR(200) NULL AFTER insurance_company_id,
 ADD COLUMN patient_payable DECIMAL(10,2) DEFAULT 0.00 AFTER pending_amount,
 ADD COLUMN created_by VARCHAR(50) NULL AFTER remarks,

```

```

 ADD COLUMN updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
 ON UPDATE CURRENT_TIMESTAMP AFTER created_at;
...

```

---

### ## 3. COMPLETE DATA FLOW

#### ### FLOW 1 – Patient Admission to Billing Creation

...

STEP 1: Receptionist creates new admission

```

↓
ipd_admissions → INSERT 1 row
hospital_beds → UPDATE bed_status = 'Occupied'

```

STEP 2: System auto-creates billing master

```

↓
ipd_billing_master → INSERT 1 row
bill_id = generateBillId() → BILL-20250726-0001
admission_id = ADM-20250726-0001
patient_id = PAT-001
doctor_id = D005
admission_date = 2025-07-20
all charges = 0.00
grand_total = 0.00
amount_paid = 0.00
balance_due = 0.00
payment_status = 'Pending'
billing_status = 'OPEN'

```

STEP 3: Receptionist collects advance (optional)

```

↓
→ Go to FLOW 3 (Payment Recording)

```

...

---

#### ### FLOW 2 – Room Rent Generation (Daily)

...

STEP 1: Receptionist clicks "Generate Room Rent"

Selects date range: From 2025-07-20 To 2025-07-24

↓

STEP 2: System fetches bed details

```

SELECT amount_per_day, nursig_charge, doctor_charge, total_bed_amount,
 ward_name, room_name, bed_number
FROM hospital_beds
WHERE sl_no = (SELECT bed_no FROM ipd_admissions WHERE admission_id = ?)

```

↓

Result: amount\_per\_day=1500, nursig\_charge=300, doctor\_charge=200,  
total=2000

↓

STEP 3: System shows PREVIEW to receptionist:

Date	Bed Rent	Nursing	Duty Dr	Total
20-Jul	1,500	300	200	2,000
21-Jul	1,500	300	200	2,000

22-Jul	1,500	300	200	2,000
23-Jul	1,500	300	200	2,000
24-Jul	1,500	300	200	2,000
TOTAL	7,500	1,500	1,000	10,000

[Cancel] [Confirm & Add]

↓

STEP 4: Receptionist confirms → System INSERTs 5 rows in ipd\_billing\_items  
(1 row per day – each with bed\_rent, nursing\_charge, duty\_dr\_charge)

↓

STEP 5: System calls recalculateMaster(bill\_id)  
→ room\_charges updated to ₹10,000  
→ subtotal updated  
→ grand\_total updated  
→ balance\_due updated  
→ ipd\_billing\_master updated instantly  
...

---

### FLOW 3 – Recording Any Charge (Lab, Doctor, Pharmacy, etc.)

...

STEP 1: Receptionist clicks "+ Add Charge"

Modal opens with:

- Charge Date (default: today)
- Charge Type (dropdown: LAB, DOCTOR\_VISIT, PHARMACY, etc.)
- Department (optional)
- Description (item/service name)
- Quantity
- Unit Price (₹)
- Discount (₹) (optional)
- Notes/items\_json data (optional)

↓

STEP 2: System calculates:  
total\_amount = (quantity × unit\_price) - discount\_amt

↓

STEP 3: INSERT 1 row into ipd\_billing\_items

↓

STEP 4: System calls recalculateMaster(bill\_id)  
→ Relevant category column updated (e.g. lab\_charges += 500)  
→ subtotal updated  
→ grand\_total updated  
→ balance\_due updated  
→ ipd\_billing\_master updated instantly

↓

STEP 5: UI refreshes – financial summary panel shows new totals  
...

---

### FLOW 4 – Recording a Payment

...

STEP 1: Receptionist clicks "+ Record Payment"

Modal shows:

- Current Balance Due: ₹15,000
- Payment Date

```

- Payment Type (ADVANCE / PARTIAL / FINAL / REFUND)
- Payment Mode (CASH / UPI / CARD / BANK / CHEQUE)
- Amount (₹)
- Reference No (optional – for UPI, cheque etc.)
- Remarks (optional)
↓
STEP 2: Validation:
- Amount > 0
- Payment date >= admission date
- For FINAL: warn if amount < balance_due
↓
STEP 3: INSERT 1 row into ipd_payment
is_insurance = 0 (this is patient payment)
↓
STEP 4: System calls recalculateMaster(bill_id):
amount_paid = SUM(amount FROM ipd_payment
WHERE bill_id AND is_insurance=0 AND payment_type !=
'REFUND')
refund_total = SUM(amount FROM ipd_payment WHERE bill_id AND
payment_type='REFUND')
amount_paid = amount_paid - refund_total
balance_due = grand_total - amount_paid - insurance_received_amount
advance_amount = SUM(amount WHERE payment_type='ADVANCE' AND
is_insurance=0)
→ Determine payment_status:
IF amount_paid == 0 → 'Pending'
ELSE IF balance_due <= 0 → 'Paid'
ELSE → 'Partial'
→ UPDATE ipd_billing_master
↓
STEP 5: UI refreshes – payment history table + financial summary updated
` ``

FLOW 5 – Insurance Patient Flow
...

STEP 1: When admission is created OR anytime during stay:
Receptionist clicks "Set Insurance Details"
Modal:
- Insurance Type: INSURANCE / CORPORATE
- Company Name
- Policy Number
- Claim Number
- Approval Number
- Approved Amount (₹)
↓
STEP 2: INSERT/UPDATE 1 row in ipd_insurance
pending_amount = approved_amount - received_amount (starts at 0)
patient_payable = grand_total - approved_amount
↓
STEP 3: System updates ipd_billing_master:
bill_type = 'INSURANCE' or 'CORPORATE'
insurance_approved_amount = ipd_insurance.approved_amount
patient_payable = grand_total - insurance_approved_amount
→ calls recalculateMaster()
↓
STEP 4: When insurance company sends payment:

```

```

 Receptionist clicks "+ Insurance Receipt"
 Modal:
 - Amount Received (₹)
 - Reference No (claim settlement ref)
 - Date
 ↓
STEP 5: INSERT 1 row into ipd_payment
 is_insurance = 1 ← insurance payment flag
 payment_type = 'FINAL'
 payment_mode = 'INSURANCE'
 ↓
STEP 6: UPDATE ipd_insurance:
 received_amount += new amount
 pending_amount = approved_amount - received_amount
 ↓
STEP 7: recalculateMaster():
 insurance_received_amount = SUM(amount FROM ipd_payment WHERE
is_insurance=1)
 balance_due = grand_total - amount_paid - insurance_received_amount
 → If balance_due <= 0 → payment_status = 'Paid'
...

FLOW 6 – Discharge & Finalization
...

STEP 1: Doctor gives discharge order
 Receptionist opens Discharge form
 ↓
STEP 2: System checks billing:
 IF balance_due > 0:
 → Show warning: "Outstanding balance ₹X,XXX remaining"
 → Show option: "Collect payment before discharge"
 ↓
STEP 3: Generate Final Bill (before discharging)
 → Room rent auto-generated for any missing days
 → All charges finalized
 ↓
STEP 4: Collect remaining balance:
 → Record FINAL payment via FLOW 4
 ↓
STEP 5: Set Discharge Date
 ipd_admissions → UPDATE discharge_date, status = 'Discharged'
 hospital_beds → UPDATE bed_status = 'Available'
 ↓
STEP 6: Update billing master:
 discharge_date = [date]
 total_days = DATEDIFF(discharge_date, admission_date)
 billing_status = 'FINALIZED'
 → recalculateMaster()
 ↓
STEP 7: Print Final Bill + Receipt
...

4. THE `recalculateMaster()` ALGORITHM

```



This is the HEART of the system. Called after every item add, payment, or discount change.

...

recalculateMaster(bill\_id):

```
— STEP 1: Sum billing items by charge_type —
SELECT charge_type, SUM(total_amount) as cat_total
FROM ipd_billing_items
WHERE bill_id = ? AND status != 'CANCELLED'
GROUP BY charge_type
```

Map results:

```
ROOM_RENT → room_charges
DOCTOR_VISIT → doctor_charges
LAB → lab_charges
RADIOLOGY → radiology_charges
PHARMACY → pharmacy_charges
OT → ot_charges
PROCEDURE → procedure_charges
CONSUMABLE → consumable_charges
OTHER → other_charges
```

```
— STEP 2: Calculate subtotal —
subtotal = room + doctor + lab + radiology + pharmacy + ot + procedure +
consumable + other
```

```
— STEP 3: Apply discount —
IF discount_percentage > 0:
 discount_amount = subtotal × (discount_percentage / 100)
grand_total = subtotal - discount_amount
```

```
— STEP 4: Calculate payments from ipd_payment —
SELECT
 SUM(CASE WHEN is_insurance=0 AND payment_type != 'REFUND' THEN amount ELSE 0
END) as paid,
 SUM(CASE WHEN payment_type='REFUND' THEN amount ELSE 0 END)
as refunded,
 SUM(CASE WHEN is_insurance=1 THEN amount ELSE 0 END)
as ins_rcvd,
 SUM(CASE WHEN payment_type='ADVANCE' AND is_insurance=0 THEN amount ELSE 0 END)
as advance
FROM ipd_payment
WHERE bill_id = ?

amount_paid = paid - refunded
insurance_received_amount= ins_rcvd
advance_amount = advance
```

```
— STEP 5: Insurance —
SELECT approved_amount FROM ipd_insurance WHERE bill_id = ?
insurance_approved_amount = approved_amount (or 0)
patient_payable = grand_total - insurance_approved_amount
```

```
— STEP 6: Balance —
balance_due = grand_total - amount_paid - insurance_received_amount
```

```
— STEP 7: Payment Status —
IF amount_paid == 0 AND insurance_received_amount == 0:
```

```

 payment_status = 'Pending'
ELSE IF balance_due <= 0:
 payment_status = 'Paid'
ELSE:
 payment_status = 'Partial'

```

— STEP 8: UPDATE ipd\_billing\_master —

```

UPDATE ipd_billing_master SET
 room_charges = ?,
 doctor_charges = ?,
 lab_charges = ?,
 radiology_charges = ?,
 pharmacy_charges = ?,
 ot_charges = ?,
 procedure_charges = ?,
 consumable_charges = ?,
 other_charges = ?,
 subtotal = ?,
 grand_total = ?,
 insurance_approved_amount = ?,
 insurance_received_amount = ?,
 patient_payable = ?,
 advance_amount = ?,
 amount_paid = ?,
 balance_due = ?,
 payment_status = ?,
 updated_by = ?,
 updated_at = NOW()
WHERE bill_id = ?

```

---

## ## 5. FINANCIAL SUMMARY EXAMPLES

### ### Example 1 – Self Patient, No Payment Yet

```

Grand Total = ₹20,000
Amount Paid = ₹0
Balance Due = ₹20,000
Payment Status = PENDING

```

### ### Example 2 – Self Patient, Advance Only

```

Grand Total = ₹20,000
Advance Paid = ₹5,000
Amount Paid = ₹5,000
Balance Due = ₹15,000
Payment Status = PARTIAL

```

### ### Example 3 – Self Patient, Multiple Payments

```

Grand Total = ₹30,000
Advance = ₹5,000
Partial = ₹8,000
Partial = ₹2,000
Amount Paid = ₹15,000

```

```
Balance Due = ₹15,000
Payment Status = PARTIAL
\\`
```

### ### Example 4 – Insurance Patient (Fully Settled)

```
Grand Total = ₹80,000
Insurance Approved = ₹60,000
Patient Payable = ₹20,000 (80k - 60k)
Patient Paid = ₹20,000
Insurance Received = ₹60,000
Balance Due = ₹0 (80k - 20k - 60k)
Payment Status = PAID
\\`
```

### ### Example 5 – Discount Applied

```
Subtotal = ₹25,000
Discount (10%) = ₹2,500
Grand Total = ₹22,500
Amount Paid = ₹10,000
Balance Due = ₹12,500
Payment Status = PARTIAL
\\`
```

---

## ## 6. BILLING STATUS LIFECYCLE

```
\\`
OPEN
|
↓ (patient starts treatment)
UNDER_TREATMENT
|
↓ (doctor gives discharge order)
DISCHARGE_PENDING
|
↓ (balance cleared + final bill printed)
FINALIZED
|
(or at any point)
↓
CANCELLED
\\`
```

---

## ## 7. MVC FILE STRUCTURE

```
\\`
ipd_management/
|
├── models/
| ├── IpdBillingMaster.php ← NEW (manages ipd_billing_master)
| ├── IpdBillingItem.php ← NEW (manages ipd_billing_items)
| ├── IpdPayment.php ← NEW (manages ipd_payment)
| └── IpdInsurance.php ← NEW (manages ipd_insurance)
```

```

├── Admission.php ← UNTOUCHED
├── IpdBilling.php ← UNTOUCHED (old, keep)
├── Payment.php ← UNTOUCHED (old, keep)
├── controllers/
│ ├── IpdBillingMasterController.php ← NEW
│ ├── IpdBillingItemController.php ← NEW
│ ├── IpdPaymentController.php ← NEW
│ ├── IpdInsuranceController.php ← NEW
│ ├── IpdBillingController.php ← UNTOUCHED (old, keep)
│ └── PaymentsController.php ← UNTOUCHED (old, keep)
├── views/
│ ├── billing/
│ │ ├── index.php ← REWRITE (the billing terminal)
│ │ ├── print_final.php ← NEW
│ │ ├── print_receipt.php ← NEW
│ │ ├── print_interim.php ← NEW
│ │ └── print_insurance.php ← NEW
│ ├── admissions/
│ │ └── index.php ← MODIFY (add billing button)
│ └── discharge/
│ └── index.php ← MODIFY (add balance check)
├── routes/
│ └── api.php ← MODIFY (add 4 new routes)
├── public/
│ ├── assets/
│ │ ├── js/
│ │ │ └── ipd_billing_terminal.js ← NEW
│ │ └── css/
│ │ └── ipd_billing_terminal.css ← NEW
│ └── ...
└── ...

```

---

## ## 8. API ENDPOINTS

### ### Billing Master

Method	Endpoint	Action	Description
GET	`api/ipd-billing-master?admission_id=ADM-xxx`	getOrCreate	Load or create billing master
GET	`api/ipd-billing-master?bill_id=BILL-xxx`	get	Get full billing summary
POST	`api/ipd-billing-master`	action=discount	applyDiscount   Set discount amount/%
POST	`api/ipd-billing-master`	action=status	updateStatus   Change billing status
POST	`api/ipd-billing-master`	action=bill_type	updateBillType   Switch SELF/INSURANCE

### ### Billing Items

Method	Endpoint	Action	Description
GET	`api/ipd-billing-items?bill_id=BILL-xxx`	getAll	All charge rows
GET	`api/ipd-billing-items?bill_id=BILL-xxx&category=LAB`	filter	Filter by category
POST	`api/ipd-billing-items`	action=add	addItem   Add any single charge
POST	`api/ipd-billing-items`	action=room_rent	bulkRoomRent   Auto-generate room rent rows
POST	`api/ipd-billing-items`	action=cancel	cancelItem   Cancel a charge (no delete)

### ### Payments

Method	Endpoint	Action	Description
GET	`api/ipd-payment?bill_id=BILL-xxx`	getAll	Full payment history
POST	`api/ipd-payment`	action=pay	recordPayment   Record patient payment
POST	`api/ipd-payment`	action=insurance	recordInsurance   Record insurance receipt

### ### Insurance

Method	Endpoint	Action	Description
GET	`api/ipd-insurance?bill_id=BILL-xxx`	get	Get insurance details
POST	`api/ipd-insurance`	action=save	saveOrUpdate   Save/update insurance info
POST	`api/ipd-insurance`	action=received	updateReceived   Update received amount

## ## 9. BILLING TERMINAL UI LAYOUT

### ### Main Page Structure

...

IPD BILLING TERMINAL – GM HMS

Search and select admission...

← Select2 dropdown

▼ After admission is selected:

PATIENT HEADER CARD

ADM-20250726-0001

General Ward

Admitted: 20-Jul-2025

John Doe (M/45)

Room 101, Bed B-05

Discharged: –

Dr. Rajesh Kumar

Days: 6

Billing Status: [OPEN ▼]

Payment Status:  PENDING

[📄 Interim Bill]

[📄 Final Bill]

[📄 Receipt]

[📄 Insurance Claim]

LEFT PANEL (65%)

BILLING ITEMS

RIGHT (35%)

FINANCIAL

[+ Add Charge ▼] [ Room Rent] [Filter: All ▼]

SUMMARY

#	Date	Category	Description	Qty	₹
1	20 Jul	 ROOM RENT	Gen Ward-B05	1	2000
2	21 Jul	 ROOM RENT	Gen Ward-B05	1	2000
3	21 Jul	 LAB	CBC	1	500
4	22 Jul	 DOCTOR	Cardio Visit	1	1500
5	23 Jul	 PHARMACY	Paracetamol	10	50

[← Prev] Page 1 of 1 [Next →]

#### PAYMENT HISTORY

[+ Record Payment] [+ Insurance Receipt]

#	Date	Type	Mode	Amount	Ref No
1	20 Jul	ADVANCE	CASH	₹5,000	—
2	22 Jul	PARTIAL	UPI	₹3,000	UPI-123

Room Rent  
₹ 12,000

Doctor Visit

₹ 1,500

Laboratory

₹ 500

Radiology

₹ 0

Pharmacy

₹ 500

OT / Procedure

₹ 0

Consumables

₹ 0

Miscellaneous

₹ 0

Subtotal

₹ 14,500

Discount (-10%)

-₹ 1,450

GRAND TOTAL

₹ 13,050

Ins. Approved

₹ 0

Ins. Received

	₹ 0
	Patient Payable
	₹ 13,050
	Total Paid
	₹ 8,000
	BALANCE DUE
	₹ 5,050
	[Apply
Discount]	[Insurance
Info]	

...

---

## 10. ALL MODALS (7 total)

### Modal 1 - Add Charge

Add Billing Charge	
Charge Date	[2025-07-26 ▼]
Charge Type	[Select Category ▼] (ROOM_RENT / DOCTOR_VISIT / LAB / RADIOLOGY / PHARMACY / OT / PROCEDURE / CONSUMABLE / OTHER)
Department	[Optional]
Description	[Item / Service Name]
Quantity	[1] Unit Price ₹ [0.00]
Discount ₹	[0.00]
Total Amount: ₹ 0.00 (auto-calculated)	
Notes	[Optional]
[Cancel] [Add Charge]	

...

### Modal 2 - Generate Room Rent

Generate Room Rent				
Room: General Ward - Room 101 - Bed B-05				
Daily Rate: Bed ₹1,500 + Nursing ₹300 + Dr ₹200 = ₹2,000				
From Date [2025-07-20]		To Date [2025-07-25]		
PREVIEW:				
Date	Bed	Nurse	Duty	Total
20-Jul	1500	300	200	2,000
21-Jul	1500	300	200	2,000

... 6 days		12,000
------------	--	--------

⚠ Already generated dates will be skipped  
 [Cancel] [Confirm & Generate]

...

### ### Modal 3 - Record Payment

...

Record Patient Payment	
Current Balance Due: ₹15,000	
Payment Date	[2025-07-26]
Payment Type	[ADVANCE / PARTIAL / FINAL / REFUND ▼]
Payment Mode	[CASH / UPI / CARD / BANK / CHEQUE ▼]
Amount ₹	[0.00]
Reference No	[Optional - UPI txn, cheque no]
Remarks	[Optional]
Balance after payment: ₹ 15,000 (live preview)	
[Cancel] [Record Payment]	

...

### ### Modal 4 - Insurance Receipt

...

Record Insurance Receipt	
Insurance: Star Health - Policy: SH-2025-001	
Approved Amount: ₹60,000	
Already Received: ₹0	
Pending Claim: ₹60,000	
Receipt Date	[2025-07-26]
Amount ₹	[0.00]
Reference No	[Claim settlement ref]
Remarks	[Optional]
[Cancel] [Record Receipt]	

...

### ### Modal 5 - Apply Discount

...

Apply Discount	
Current Subtotal: ₹24,000	
Discount Amount ₹	[0.00] ← updates % field
OR	
Discount %	[0.00] ← updates ₹ field
Grand Total after discount: ₹24,000 (live preview)	
[Cancel] [Apply Discount]	

...

### ### Modal 6 - Insurance Details

...

Insurance / Corporate Details	
Type	[INSURANCE / CORPORATE ▼]



Company Name	[Star Health / TCS etc.]
Company ID	[Optional - internal ref]
Policy Number	[SH-2025-001]
Claim Number	[Optional]
Approval Number	[Optional]
Approved Amount	₹ [0.00]
Remarks	[Optional]
	[Cancel] [Save Details]

...

### ### Modal 7 - Change Billing Status

...

Change Billing Status	
Current Status: OPEN	
New Status	[OPEN ▼] OPEN UNDER_TREATMENT DISCHARGE_PENDING FINALIZED CANCELLED
Reason	[Optional note about status change] [Cancel] [Update Status]

...

---

## ## 11. PRINT TEMPLATES (4 files)

### ### `print\_final.php` - Final Discharge Bill

...

#### Content:

- Hospital header (logo, name, address)
- Patient info: Name, Age, Sex, Ward, Bed, Admission Date, Discharge Date, Days
- Doctor: Name, Specialization
- Bill ID, Bill Date
- ITEMIZED CHARGES TABLE (all rows from ipd\_billing\_items, grouped by category)
- Category subtotals
- Discount applied
- Grand Total
- PAYMENT HISTORY table (all rows from ipd\_payment)
- Balance Due
- Insurance details (if applicable)
- Signature lines: Cashier / Patient / Doctor

...

### ### `print\_receipt.php` - Payment Receipt

...

#### Content:

- Receipt Number (PAY-YYYYMMDD-XXXX)
- Date & Time
- Patient Name, Admission ID
- Payment Type, Mode, Amount
- Reference No
- Balance remaining
- Cashier name

...

### `print\_interim.php` - Interim / Daily Bill

...

Content:

- "INTERIM BILL" stamp
- Patient info
- Charges up to today (not final)
- Current subtotal, discount, grand total
- Amount paid so far, balance due
- "Final amount may change upon discharge" note

...

### `print\_insurance.php` - Insurance Claim Statement

...

Content:

- Patient info
- Insurance Company, Policy No, Claim No, Approval No
- Full itemized charges (sorted by category for insurer review)
- Grand Total
- Insurance Approved Amount
- Patient Payable Amount
- Insurance Received to date
- Pending Claim Amount
- Hospital stamp/seal

...

---

## 12. EXECUTION ORDER (When You Say "Start Coding")

...

Phase 1 - Database (Run SQL first)

- Step 1.1 ALTER ipd\_billing\_master (ADD 7 columns)
- Step 1.2 ALTER ipd\_billing\_items (ADD 6 columns)
- Step 1.3 ALTER ipd\_payment (ADD 1 column)
- Step 1.4 ALTER ipd\_insurance (ADD 5 columns)
- Step 1.5 Verify charge\_type ENUM has all 9 values - ALTER if needed

Phase 2 - Models (PHP backend)

- Step 2.1 IpdBillingMaster.php (with recalculateMaster)
- Step 2.2 IpdBillingItem.php (with room rent generation)
- Step 2.3 IpdPayment.php (always INSERT)
- Step 2.4 IpdInsurance.php (save or update)

Phase 3 - Controllers + Routes

- Step 3.1 IpdBillingMasterController.php
- Step 3.2 IpdBillingItemController.php
- Step 3.3 IpdPaymentController.php
- Step 3.4 IpdInsuranceController.php
- Step 3.5 Update routes/api.php

Phase 4 - Frontend

- Step 4.1 ipd\_billing\_terminal.css
- Step 4.2 views/billing/index.php (main terminal page)
- Step 4.3 ipd\_billing\_terminal.js (all API calls + UI logic)

Phase 5 - Print Templates

- Step 5.1 print\_final.php

Step 5.2 print\_receipt.php  
Step 5.3 print\_interim.php  
Step 5.4 print\_insurance.php

Phase 6 – Integration

Step 6.1 views/admissions/index.php (add billing button)  
Step 6.2 views/discharge/index.php (add balance check)

Phase 7 – Testing

Step 7.1 Test SELF patient full flow  
Step 7.2 Test INSURANCE patient full flow  
Step 7.3 Test room rent auto-generation  
Step 7.4 Test all 4 print templates  
```

13. ONE QUESTION REMAINING

> Before coding starts, please run this query and share the result:
> ```sql
> SHOW COLUMNS FROM ipd_billing_items LIKE 'charge_type';
> ```
> Need to confirm the full ENUM list (ROOM_RENT, DOCTOR_VISIT, LAB, RADIOLOGY,
PHARMACY, OT, PROCEDURE, CONSUMABLE, OTHER).
> If any are missing, one more ALTER is needed.

Everything else is 100% clear and ready to build.